

Hodge-relative completions of characteristic brane couplings

Josiah Calvert

Abstract

Chern–Simons theory is sometimes described, loosely, as Yang–Mills with the metric removed: the characteristic form is transgressed one degree down and its Hodge star is discarded in favor of a locally dynamical quantity. We ask what happens if we reverse this step: given a closed, gauge-invariant *characteristic* form on a D-brane worldvolume, does putting the metric back in – forming its Hodge square – ever produce a novel local coupling, not already present in the Dirac–Born–Infeld (DBI) or Wess–Zumino (WZ) sectors? We show that, to first order in a change of representative, it is precisely the codifferential $\delta_g P$ that obstructs such a coupling; representative-independence at this order holds exactly on the co-closed locus. We present this as a necessary and sufficient condition – with an explicit formula for the obstruction in terms of that codifferential – and apply the resulting bookkeeping to the D4-brane at leading order, proving a small, honest no-go: every coupling built from Hodge-completed field content at this order belongs to a known sector – DBI, ordinary flux kinetic terms, or characteristic descent – and the one pairing that would mix the two generators fails for form-degree reasons. We close by identifying, through the same bookkeeping, the first point at which brane extrinsic geometry could produce a mixed class, and this as a precise open problem.

1 Introduction

There is a familiar sense in which Chern–Simons theory is ‘Yang–Mills without a metric.’ Given a gauge-invariant characteristic polynomial $P(F)$, closed by the Bianchi identity, with local transgression $P(F) = dQ_{\text{CS}}(A)$, the functional $\int Q_{\text{CS}}(A)$ never calls on a Riemannian structure: it is built from A , F , and the wedge product alone. The Yang–Mills functional $\int \text{Tr}(F \wedge *_g F)$, by contrast, exists only once a metric has been chosen, and it is exactly that dependence which turns a quantity sensitive only to a bundle’s characteristic classes into a genuinely local field theory, with a stress tensor, propagation, and scattering. We take this as a slogan rather than a theorem: the two functionals are related by transgression *and* by deleting a metric, two logically separate operations, and we are careful below not to conflate them.

The question which we in fact ask is the inverse one, stated without reference to a particular pair of actions: *given a closed, gauge-invariant form built from the characteristic data of a D-brane – curvatures, fluxes, Ramond–Ramond field strengths – when does reinstating a worldvolume metric, by taking its Hodge square, give a local coupling not accounted for by the standard DBI and WZ sectors?* A first guess is that this should be controlled by de Rham cohomology: forms in the same class should complete to the same local theory. This guess is wrong, in fact its failure is not a technical nuisance but the entire content of the problem: a

fixed cohomology class only remembers global, bundle-level information while its Hodge square is manifestly a function of the chosen representative, the gap between the two precisely being where new local physics could live.

Making this precise requires some care, and there are three clarifications worth stating before seeing any formula. First, the relevant ambiguity in a representative is *not* a gauge transformation of a connection: for $F = dA$, shifting A by an exact one-form leaves F untouched so there is no room here for new physics. The ambiguity we study instead arises for truly closed characteristic forms such as the Neveu–Schwarz field strength $H = dB$, whose cohomology class is fixed but whose specific representative – the choice of *curving* on the underlying gerbe – is extra, physical data not captured by $[H]$ alone; this is precisely the content of differential cohomology [4, 5], and is in fact the correct and rigorous setting for the question we ask. Second, ‘topological’ has to be narrowly defined: we reserve it for forms that are exact *within the characteristic algebra itself*, in the spirit of Chern–Simons descent, rather than for the much broader and less informative class of forms with vanishing Euler–Lagrange variation. Third, we make no claim that any of the individual algebraic ingredients below are new; the variational bicomplex and local BRST cohomology already supply the correct home for local Lagrangians modulo total derivatives and gauge redundancy [1, 2], and the classification of higher-derivative D-brane operators, including

terms quadratic in H , is already extensive and tightly constrained by T-duality [8, 7, 6, 9]. What has not, to our knowledge, been isolated is the precise criterion governing when a metric completion of characteristic data is representative-independent, and the use of that criterion as a classification tool.

Our results are a pair of small, self-contained statements. Theorem 3.2 identifies that criterion: to first order in a change of representative, the obstruction is precisely the codifferential $\delta_g P$, and representative-independence at this order holds iff the representative is co-closed. Theorem 4.2 then applies this bookkeeping to the D4-brane and shows that its entire leading-order Hodge-completed sector, with no exceptions, already belongs to DBI, ordinary flux dynamics, or characteristic descent; the one candidate which would genuinely mix the two available generators does not exist, for reasons of form degree alone. This is a calibration result: it says exactly how far the obvious search space is from producing anything new, and Section 5 uses the same bookkeeping to say exactly where a non-trivial class could first appear.

2 The relative completion problem

Fix an oriented worldvolume W of dimension d with metric g , and a field bundle $E \rightarrow W$ carrying the variational bicomplex splitting $d = d_H + d_V$ on its infinite jet bundle [1]. A local Lagrangian is a horizontal top form $L \in \Omega_{\text{loc}}^{d,0}(J^\infty E)$, with Euler–Lagrange form defined through $d_V L = \text{EL}(L) + d_H \Theta_L$.

Definition 2.1. *Let*

$$\mathcal{C}_{\text{ext}} = \langle \phi^* \Omega^\bullet(X), \mathcal{F}, H, R, D \rangle_{\wedge, d, D}$$

be the algebra of gauge-invariant forms built covariantly from the pulled-back bulk fields and the worldvolume field strength, using no metric. Its descent sector is

$$\mathcal{W}^d = \{ \tau \in \mathcal{C}_{\text{ext}}^d : \tau = d\sigma, \sigma \in \mathcal{C}_{\text{ext}}^{d-1} \},$$

*the elements exact within $\mathcal{C}_{\text{ext}}$ itself – the natural home of Chern–Simons/Wess–Zumino descent. The metric completion is $\mathcal{C}_H(g) = \langle \mathcal{C}_{\text{ext}}, *_g \rangle_{\wedge, d, D, *_g}$, obtained by adjoining the Hodge star.*

Definition 2.2. *The Hodge-relative variational coupling space in degree d is*

$$\mathfrak{V}_{H/\text{Top}}^d(g) = \frac{\mathcal{C}_H^d(g)}{\ker(\text{EL}) + \mathcal{W}^d + \mathcal{R}^d}, \quad (1)$$

where $\mathcal{R}^d$ collects the standard redundancies of the effective description: Bianchi and algebraic identities, integration by parts, chosen field redefinitions. By

the triviality theorem for the variational bicomplex, $\ker(\text{EL})$ coincides locally with $d_H \Omega_{\text{loc}}^{d-1,0}$ [1], so this recovers the expected statement that boundary terms are discarded.

Figure 1 depicts the quotient of Definition 2.2. A nonzero class in $\mathfrak{V}_{H/\text{Top}}^d(g)$ is, by construction, a local coupling with nonvanishing Euler–Lagrange variation that is neither a boundary term nor a characteristic/WZ coupling nor removable by the identities in $\mathcal{R}^d$: exactly the sense in which a Hodge completion can fail to be redundant.

3 The completion theorem

Fix a class $[P] \in H_{\text{dR}}^p(W; \mathbb{R})$ represented by a closed form $P \in \mathcal{C}_{\text{ext}}^p$, $dP = 0$. As discussed in the Introduction, we do *not* interpret a shift of representative as a gauge transformation of an underlying potential; we interpret it as the choice of differential-cohomological refinement – the curving – of a fixed class, which for the Neveu–Schwarz flux is exactly the freedom $H \mapsto H + d\mu$ by a globally defined two-form μ , corresponding to a different choice of B -field profile with the same periods [4, 5].

Definition 3.1. *For $P \in \mathcal{C}_{\text{ext}}^p$, the Hodge completion of P is $K_g(P) = P \wedge *_g P$. Since $*_g P \in \Omega^{d-p}$, this is always a top-degree form, $K_g(P) \in \Omega^d(W)$, regardless of p .*

Theorem 3.2 (Completion criterion). *Let W be closed and oriented, and let δ_g denote the codifferential of g . For any $(p-1)$ -form μ ,*

$$K_g(P + d\mu) - K_g(P) \equiv 2\varepsilon \mu \wedge *_g \delta_g P + d\mu \wedge *_g d\mu \quad (2)$$

modulo d_H -exact terms, for a sign $\varepsilon = \varepsilon(p, d) = \pm 1$ fixed once and for all by the Hodge-star and codifferential conventions in use (e.g. [3], §7). Consequently, as a class in $\mathfrak{V}_{H/\text{Top}}^d(g)$, the completion $[K_g(P)]_{\text{rel}}$ is independent of the choice of representative within $[P]$, to linear order in the shift, for every shift μ iff $\delta_g P = 0$: that is, if and only if P is co-closed, which together with $dP = 0$ means P is g -harmonic.

Proof. Expand $K_g(P + d\mu) - K_g(P) = 2d\mu \wedge *_g P + d\mu \wedge *_g d\mu$, using bilinearity of $\wedge$ and $*_g$ and the pointwise symmetry $\alpha \wedge *_g \beta = \beta \wedge *_g \alpha$ for equal-degree forms. For the linear term, Stokes’ theorem on the closed manifold W gives, as an identity of local densities modulo an exact top form,

$$d\mu \wedge *_g P \equiv (-1)^p \mu \wedge d *_g P \pmod{d(\mu \wedge *_g P)},$$

and the correction term $d(\mu \wedge *_g P)$ is exact, hence lies in $d_H \Omega_{\text{loc}}^{d-1,0} \subset \ker(\text{EL})$ and is discarded by Definition 2.2. The standard Hodge-theoretic identity relating the exterior derivative of a Hodge dual to the

$$\ker(\text{EL}) + \mathcal{W}^d + \mathcal{R}^d \hookrightarrow \mathcal{C}_H^d(g) \xrightarrow{\pi} \mathfrak{V}_{H/\text{Top}}^d(g)$$

Figure 1: The quotient defining the Hodge-relative variational coupling space: the discarded ideal (left) embeds in the completed algebra, which surjects onto the coupling space via π .

$$\begin{array}{ccc} P & \xrightarrow{K_g} & K_g(P) \\ \downarrow [\cdot]_{\text{dR}} & & \downarrow [\cdot]_{\text{rel}} \\ [P] & \dashrightarrow & [K_g(P)]_{\text{rel}} \\ & \# & \end{array}$$

Figure 2: The completion square of Theorem 3.2. The top map and both vertical quotients always exist; the dashed bottom map exists only along the harmonic locus $\delta_g P = 0$.

codifferential, $d *_g P = \varepsilon' *_g \delta_g P$ for a sign $\varepsilon' = \varepsilon'(p, d)$ fixed by convention [3], then gives Equation (2) with $\varepsilon = (-1)^p \varepsilon'$. This vanishes identically, for every choice of μ , precisely when $\delta_g P = 0$ as a form (a nonzero $\delta_g P$ can always be paired against some compactly supported μ to give a nonvanishing density, since $\mu \wedge *_g \delta_g P = \langle \mu, \delta_g P \rangle_g \text{vol}_g$ is the pointwise Hodge inner product). Together with the given closure $dP = 0$, $\delta_g P = 0$ is exactly the Hodge–de Rham harmonicity condition. $\square$

Corollary 3.3. *The completion map K_g descends, to linear order in the representative, to a well-defined function of $[P] \in H_{\text{dR}}^p(W)$ if and only if P is evaluated in the (unique, metric-dependent) harmonic representative of its class. For any other representative, the representative-dependence of K_g is entirely governed by $\delta_g P$: the failure of P to be co-closed is the obstruction.*

Figure 2 summarises the situation at hand: the top map is always available and the two vertical maps are the evident quotients, but the dashed bottom map exists – by Theorem 3.2 – only along the harmonic locus. This is the precise sense in which the Hodge star cannot turn topology into dynamics for free: it takes a representative, not a class, and the price of forgetting the representative is exactly the codifferential obstruction above. It is also, we think, a more informative statement than the vaguer assertion that ‘Hodge completion is representative-dependent’: it says exactly which representatives are safe, and exactly what controls the failure for the rest.

4 A leading-order no-go theorem on the D4-brane

We now put Definition 2.2 to work on the smallest brane for which a genuinely mixed completion is dimensionally conceivable. Let W_5 be the worldvolume of a D4-brane, $\dim W_5 = 5$, and let $\mathcal{F} = 2\pi\alpha' F + \phi^* B$ be the gauge-invariant worldvolume two-form, with $d\mathcal{F} = \phi^* H$, $H = dB$ the closed, gauge-invariant NS flux of Theorem 3.2. At the lowest order – one factor of each available closed generator, at most one Hodge insertion, no further derivatives – there are exactly four candidate top-degree densities, and their fate is settled by nothing more than form-degree arithmetic together with one elementary computation.

Lemma 4.1. $\mathcal{F} \wedge H = \frac{1}{2} d(\mathcal{F} \wedge \mathcal{F})$. In particular $\mathcal{F} \wedge H \in \mathcal{W}^5$.

Proof. Since $\mathcal{F}$ has even degree, $d\mathcal{F} \wedge \mathcal{F} = \mathcal{F} \wedge d\mathcal{F}$, so $d(\mathcal{F} \wedge \mathcal{F}) = 2d\mathcal{F} \wedge \mathcal{F} = 2\phi^* H \wedge \mathcal{F}$. The claim follows. $\square$

Lemma 4.1 deserves emphasis: it tells us that the single pairing of $\mathcal{F}$ and H which exist without any metric at all is not only in some sense quasi-topological, but is literally a total derivative, transgressed from $\frac{1}{2}\mathcal{F} \wedge \mathcal{F}$ in precisely the way a Chern–Simons density is transgressed from a characteristic polynomial. In short, it is something that has been spoken for already.

Theorem 4.2 (Leading-order vanishing). *Within the leading-order ansatz consisting of one factor of each generator, at most one Hodge insertion, and no additional derivatives, the only degree-five monomials are $\mathcal{F} \wedge H$, $\mathcal{F} \wedge *_5 \mathcal{F}$, and $H \wedge *_5 H$. These lie respectively in $\mathcal{W}^5$, the DBI kinetic sector, and the ordinary flux-kinetic sector. There is no fourth, genuinely mixed class: the only pairing that would combine one factor of each generator with a single Hodge insertion, $\mathcal{F} \wedge *_5 H$, has degree $2 + 2 = 4 \neq 5$ and simply does not define a Lagrangian density on W_5 . Consequently*

$$\mathfrak{V}_{H/\text{Top}}^5(g) = 0$$

at this order: the naive characteristic-to-Hodge completion problem produces no new interaction on the D4-brane.

Proof. Immediate from Table 1: $\mathcal{F} \wedge H \in \mathcal{W}^5$ by Lemma 4.1, and $\mathcal{F} \wedge *_5 \mathcal{F}$, $H \wedge *_5 H$ are, by inspection in local coordinates, proportional to $F_{\mu\nu} F^{\mu\nu} \text{vol}_g$ and

Candidate	Degree	Star insertions	Fate
$\mathcal{F} \wedge H$	$2 + 3 = 5$	0	$\mathcal{W}^5$ (Lemma 4.1)
$\mathcal{F} \wedge *_5 \mathcal{F}$	$2 + 3 = 5$	1	DBI kinetic term
$H \wedge *_5 H$	$3 + 2 = 5$	1	ordinary flux-kinetic term [8]
$\mathcal{F} \wedge *_5 H$	$2 + 2 = 4$	1	<i>not a top form</i>

Table 1: The complete leading-order candidate list on the D4-brane.

$H_{\mu\nu\rho}H^{\mu\nu\rho}\text{vol}_g$ respectively – the standard quadratic DBI term and an ordinary flux-kinetic term already present in the higher-derivative D-brane literature [8]. Since $*_5$ sends p -forms to $(5 - p)$ -forms, no other top-degree pairing of $\mathcal{F}$ and H with a single star exists. $\square$

The value of this result is therefore diagnostic rather than constructive: it shows that the simplest intrinsic D4 search space is exhausted before genuinely mixed dynamics can appear, with the emptiness traceable to nothing more exotic than form-degree bookkeeping. A search that had instead reported $H \wedge *_5 H$ as a new coupling would have been reporting a term already classified by T-duality analyses of higher-derivative brane actions [8]; the actual content of Theorem 4.2 is the classification, not any single entry in it.

5 Where new physics could first appear

Theorem 4.2 has an unfortunate limitation in that the search only used intrinsic worldvolume data, namely $\mathcal{F}$ and H . But a D-brane is not only a manifold carrying fields. It is an embedded submanifold, and in particular the geometry contains information that cannot be recovered from the intrinsic metric and fluxes alone. The Wess–Zumino sector already knows of the tangent and normal bundles through R_T and R_N , while the embedding itself is measured by the second fundamental form Ω , a one-form valued in the normal bundle.

Such considerations suggest that the construction should naturally be enlarged. The curvature forms R_T and R_N introduce new characteristic data, while Ω provides the first genuinely extrinsic object to which that data can interact. The distinction matters because the obstruction found above was ultimately a degree-counting obstruction: with only $\mathcal{F}$ and H , the Hodge completion cannot produce a new mixed local density at leading order. Once the embedding geometry is included, there are new ways to convert transverse information into worldvolume data.

The first candidates for these therefore arise from combining characteristic curvature with the extrinsic geometry. A four-form such as

$$\text{Tr}(R_T \wedge R_T)$$

could be combined with a one-form constructed using Ω and the intrinsic fluxes. Schematically, one may look for terms of the form

$$\text{Tr}(R_T \wedge R_T) \wedge \iota_\Omega(\mathcal{F} \text{ or } H)$$

or similar contractions involving the normal curvature. The point is not that this particular expression is known to survive, but rather to illustrate the new possibility: characteristic information from the tangent or normal bundle can now interact with the local geometry of the embedding.

There is a further, more geometric possibility. Rather than treating Ω simply as an additional generator in the characteristic expansion, one can use it to turn transverse data into an intrinsic tensor before applying the Hodge pairing. In so doing the embedding itself becomes part of the map from closed background to local worldvolume dynamics. The resulting invariant need not be a characteristic form or an ordinary flux kinetic term, and its behaviour can be dependant on the geometry of the embedding rather than only the intrinsic curvature.

This is the point at which the original obstruction can genuinely change. The intrinsic calculation showed that the Hodge construction has no place to produce a new D4 coupling at leading order. Adding Ω does not evade the result, and rather changes the space in which the question is being asked. The interesting question is then whether the enlarged construction produces a nonzero, covariant local invariant, and whether that invariant has a geometric interpretation rather than simply reducing to some already familiar curvature or flux term.

Open Problem 5.1. *Extend $\mathcal{C}_{\text{ext}}$ to include the tangent and normal curvatures R_T, R_N together with the second fundamental form Ω . Determine the minimal total derivative order n_* at which $\mathfrak{V}^5 * H / \text{Top}(g)$ contains a class that is not reducible to DBI, ordinary curvature- or flux-kinetic terms, or $\mathcal{W}^5$. Theorem 4.2 shows that $n_{**} > 0$ for the intrinsic generators $\mathcal{F}$ and H alone. The inclusion of Ω provides the first natural enlargement in which transverse background data*

can be converted into local worldvolume geometry, and consequently the first setting in which a genuinely new class can be sought.

6 Discussion

The main lesson of the construction now emerges fairly simply. A closed characteristic form contains global information, while its Hodge square depends upon the particular representative. Theorem 3.2 makes this distinction explicit: the Hodge completion is representative-independent precisely at the harmonic locus. For the D4-brane, this criterion, coupled to the degree structure of the intrinsic generators, gives a stronger conclusion. At leading order, there is simply no new intrinsic coupling to be found. The possible Hodge completions reduce to the familiar DBI or flux-kinetic terms, or remain within the characteristic descent sector, as shown in Theorem 4.2.

The natural way forward therefore is not to continue searching within the same intrinsic data, but rather to ask what changes once taking the embedding seriously. Exactly the second fundamental form Ω provides this extra structure: it carries the information about how the brane sits within the ambient space, and unlike the intrinsic curvature can couple directly to transverse background fields. This opens a different route from characteristic data to local worldvolume dynamics: no longer first form an intrinsic characteristic density and then restore the metric, but rather use the embedding geometry itself to convert transverse information into worldvolume tensors before performing the Hodge pairing.

Furthermore, this allows a useful hierarchy of questions. At linear order, does the extrinsic construction produce a nontrivial local equation of motion, or does the same harmonic obstruction reappear one level higher? At quadratic order, what combinations of the resulting extrinsic tensors are genuinely independent and which reduce to familiar worldvolume curvature invariants? Finally, on suitable embeddings can the resulting local density be given a direct geometric interpretation, with its coefficient determined by the surrounding background rather than introduced independently?

These are concrete questions rather than assumptions that an enlarged theory should produce. The intrinsic calculation establishes the first obstruction; the inclusion of the embedding geometry identifies the next possible position in which that obstacle can be tested. The purpose of the enlargement is not to assume new physics, but to determine exactly whether the geometry of the embedding can transform global background data into the genuinely local dynamics on the brane.

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